Week : 9 (26-30 October 2026)

Subject/الموضوع: Physics **Unit Title عنوان الوحدة: Space, time and motion** **Grade الصف: 11DP**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1:	I can explain the effects of friction on the motion of objects and apply frictional force concepts to solve physics problems.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	Resources: IB Physics Tsokos, ruler/tape measure, classroom activity.
Lesson 2:	I can explain Hooke's Law and use the spring constant to describe and predict the behavior of elastic materials.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	Resources: IB Physics Tsokos, ruler/tape measure, classroom activity.
Lesson 3:	I can explain buoyancy and drag forces in fluids and predict how they affect the motion of objects.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	All resources can be found in Students' Class Teams
Lesson4:	HL ONLY Learning Target: Students will be able to understand, calculate, and apply torque and the conditions for rotational equilibrium.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	All resources can be found in Students' Class Teams

Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Bring a charged calculator, Physics notebook, ruler, pencil, and coloured pens. Participate actively, complete all classwork, correct misconceptions, and upload required work to Toddle by the stated deadline.

#الأسبوع Week 2

Technology Integration دمج التكنولوجيا	Interactive simulations or video frames to visualise motion; digital graphing to plot position-time data and examine slope; calculators for speed and velocity; Toddle for resources, feedback, and submission.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: coordinates, signed quantities, scale, gradient, ratio, rearranging equations, and SI units. English: precise scientific descriptions and claim-evidence reasoning. UAE Social Studies/Design: transport systems and mobility planning.
External Assessment Integration دمج التقييمات الخارجية	DP EXTERNAL EXAM
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL Thinking: interpret and connect representations. Communication: explain graph meaning with correct vocabulary. Self-management: organise data, check units, and meet deadlines. Collaboration: discuss evidence and give respectful feedback. Wellbeing: move safely during modelling and build confidence through scaffolded challenge.
Standards المعايير	DP
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How does the choice of reference point and representation influence what we claim to know about an object's motion?
Success Criteria معايير النجاح **(Measurable)**	**Lesson1:** **Must: I must identify friction as a force that opposes relative motion between surfaces.** **Should: I should distinguish between static and kinetic friction and calculate frictional forces in simple situations.** **Could: I could analyze how friction affects the motion and energy of objects in real-world applications.** **Lesson2:** **Must: I must state Hooke's Law and identify the relationship between force and extension.** **Should: I should calculate force, extension, or spring constant using $F = kx$.** **Could: I could analyze force-extension graphs and determine when a material no longer obeys Hooke's Law.** **Lesson3:** **Must: I must identify buoyancy and drag as forces that act on objects moving in or floating in fluids.** **Should: I should explain how buoyancy and drag affect the motion and equilibrium of objects in fluids.** **Could: I could analyze situations involving floating, sinking, or terminal velocity using the principles of buoyancy and drag.** **Lesson 4:** **MUST: Define torque and identify the factors that affect the torque of a force about an axis.** **SHOULD: Calculate the torque of a force and determine whether a body is in rotational equilibrium.** **COULD: Apply the condition resultant torque = 0 to solve more complex rotational equilibrium problems involving multiple forces.**

National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء	Global Understanding العالمي التفاهم	Arabic Language العربية اللغة
	Area: UAE Transportation, Road Safety, Innovation, and Sustainable Infrastructure **المجال**		
	Implementation: Students analyse journeys on Abu Dhabi roads and public transport using reference points, position-time graphs, speed, and velocity. They connect accurate motion representation to safe speed choices, efficient route planning, smart mobility, and the UAE's development of sustainable, future-ready transport infrastructure. التطبيق		